

Overview of the Iterative Markovian Fitting

Let $\mathcal{P}$ be the space of all path measures, i.e., $\mathcal{P} = \mathcal{P}(C[0, T], \mathbb{R}^d)$, and let $\mathcal{M} \subset \mathcal{P}$ be the space of all Markovian path measures.

Let $Q \in \mathcal{M}$ be a reference measure that is associated with the following SDE:

$$dX_t = f_t(X_t)dt + \sigma_t dB_t, \quad X_0 \sim \pi_0$$

and let $\mathcal{R}(Q)$ be the reciprocal class of Q .

Let P be a Markov measure in the reciprocal class of Q , such that it satisfies the prescribed marginals, π_0, π_1 , i.e., $P_0 = \pi_0$ and $P_1 = \pi_1$. Then under assumptions on Q, π_0 , and π_1 P is unique and is equal to the Schrodinger Bridge.

Consider a sequence of path measures $(P^n)_{n \in \mathbb{N}}$ such that P^{2n+1} is a Markovian Projection of P^{2n} and P^{2n+2} is the reciprocal projection of P^{2n+1} , meaning:

$$P^{2n+1} = \text{proj}_{\mathcal{M}}(P^{2n}), \quad P^{2n+2} = \text{proj}_{\mathcal{R}(Q)}(P^{2n+1})$$

with $P^0 \in \mathcal{R}(Q)$, such that $P_0^0 = \pi_0$, and $P_1^0 = \pi_1$. Furthermore it holds that for any $n \in \mathbb{N}$ $P_0^n = \pi_0$, and $P_1^n = \pi_1$.

Under mild assumptions, the $(P^n)_{n \in \mathbb{N}}$ sequence, called the IMF sequence converges to the Schrodinger Bridge. For a clear derivation see [Diffusion Schrodinger Bridge Matching](#).

Let $P \in \mathbb{P}$ be a path measure. We denote its marginal measure as $P_{0,1}$, and its bridges as $P_{|0,1}$. With this notation we present Algorithm 1.

Assume that the families of functions $\{u_\theta\}$, and $\{u_\phi\}$, are rich enough so that they can model the optimal vector fields. Let $(\Pi^n, M^{n+1})_{n \in \mathbb{N}}$ be the optimal sequence from Algorithm 1, and let $(\bar{P}^n)_{n \in \mathbb{N}}$ be the optimal Diffusion Schrodinger Bridge sequence given by IPF. Then for any $n \in \mathbb{N}$ bigger than 1, we have $M^n = \bar{P}^n$.

In the next section we will describe the practical implementation of the algorithm in the special case where the reference measure Q is a Brownian Motion. In that case Q is associated with the following SDE:

$$dX_t = \sigma dB_t, \quad \sigma > 0$$

Furthermore, the transition kernels are simple Gaussian distribution:

$$\begin{aligned} Q_{t|0}(x_t|x_0) &= N(x_t; x_0, \sigma^2 \mathbf{I}) \\ Q_{1|t}(x_1|x_t) &= N(x_1; x_t, \sigma^2(1-t)\mathbf{I}) \end{aligned}$$

Algorithm 1:

Input: Joint distribution $\Pi_{0,1}^0$, tractable bridges $Q_{|0,1}$, number of outer iterations $N \in \mathbb{N}$

Let $\Pi^0 = \Pi_{0,1}^0 Q_{|0,1}$

for $n = 0$ **to** $N - 1$ **do**

Step 1: Solve the following minimization problem:

$$\phi^* = \operatorname{argmin}_{\phi} \left\{ \mathbb{E}_{t \sim U[0,1]} \left[\frac{1}{\sigma_t^2} \mathbb{E}_{(x_0, x_t) \sim \Pi_{0,t}^{2n}} [\|\sigma_t^2 \nabla \log Q_{t|0}(x_t|x_0) - u_{\phi}(t, x_t)\|^2] \right] \right\}$$

Step 2: Let $\Pi^{2n+1} = M_{0,1}^{2n+1} Q_{|0,1}$, where M^{2n+1} is associated by the time reversal of the following SDE:

$$dY_t = \{-f_{1-t}(Y_t) + u_{\phi^*}(1-t, Y_t)\}dt + \sigma_{1-t}dB_t, \quad Y_0 \sim \pi_0$$

Step 3: Solve the following minimization problem:

$$\theta^* = \operatorname{argmin}_{\theta} \left\{ \mathbb{E}_{t \sim U[0,1]} \left[\frac{1}{\sigma_t^2} \mathbb{E}_{(x_t, x_1) \sim \Pi_{t,1}^{2n+1}} [\|\sigma_t^2 \nabla \log Q_{1|t}(x_1|x_t) - u_{\theta}(t, x_t)\|^2] \right] \right\}$$

Step 4: Let $\Pi^{2n+2} = M_{0,1}^{2n+2} Q_{|0,1}$ where M^{2n+2} is associated by the following SDE:

$$dX_t = \{f_t(X_t) + u_{\theta^*}(t, X_t)\}dt + \sigma_t dB_t, \quad X_0 \sim \pi_0$$

Output: u_{θ^*}, u_{ϕ^*}

Algorithmic Implementation

Throughout this section we will assume that the reference measure Q , is Brownian Motion.

Step 1: Solving the Backward Regression Problem

The first step of the algorithm dictates to solve:

$$\phi^* = \operatorname{argmin}_{\phi} \left\{ \mathbb{E}_{t \sim U[0,1]} \left[\frac{1}{\sigma_t^2} \mathbb{E}_{(x_0, x_t) \sim \Pi_{0,t}^{2n}} [\|\sigma_t^2 \nabla \log Q_{t|0}(x_t|x_0) - u_{\phi}(t, x_t)\|^2] \right] \right\}$$

Because Q is Brownian Motion we can easily find the close form expression of the inner expectation by first evaluating the nabla of the logarithm of the transition kernel:

$$\begin{aligned}\nabla \log Q_{t|0}(x_t|x_0) &= -\nabla \left(\frac{\|x_t - x_0\|^2}{2\sigma^2 t} \right) \\ &= \frac{x_0 - x_t}{\sigma^2 t}\end{aligned}$$

where x_0 is sampled from the data distribution.

We prove that $x_t \sim Q_{t|0,1}(\cdot|x_0, x_1)$.

Proof. Let $h : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$ denote the argument of the inner expectation for a fixed time t , defined as:

$$h(x_0, x_t) = \left\| \frac{x_0 - x_t}{t} - u_\phi(t, x_t) \right\|^2$$

For ease of notation, we omit the outer expectation over the time distribution and denote the path measure Π^{2n} simply as Π . The optimization objective at time t thus reduces to minimizing the following quantity:

$$\mathbb{E}_{(x_0, x_t) \sim \Pi_{0,t}}[h(x_0, x_t)]$$

To reformulate this expectation as an integral over the path space $C([0, 1], \mathbb{R}^d)$, we utilize the pushforward change of variables. Let $e_{0,t} : C([0, 1], \mathbb{R}^d) \rightarrow \mathbb{R}^d \times \mathbb{R}^d$ be the evaluation mapping defined by $e_{0,t}(X) = (X_0, X_t)$. The two-time marginal $\Pi_{0,t}$ is the pushforward of the path measure Π under this mapping, i.e., $\Pi_{0,t} = (e_{0,t})_\# \Pi$. By the change of variables theorem for pushforward measures, we have:

$$\begin{aligned}\mathbb{E}_{(x_0, x_t) \sim \Pi_{0,t}}[h(x_0, x_t)] &= \int_{\mathbb{R}^d \times \mathbb{R}^d} h(x_0, x_t) \Pi_{0,t}(\mathrm{d}x_0, \mathrm{d}x_t) \\ &= \int_{C([0, 1], \mathbb{R}^d)} h(X_0, X_t) \Pi(\mathrm{d}X) \\ &= \mathbb{E}_{X \sim \Pi}[h(X_0, X_t)]\end{aligned}\tag{1}$$

Let $G = \sigma(X_0, X_1)$ be the sigma algebra of the endpoints (X_0, X_1) , then from the tower property of conditional expectation we have:

$$\mathbb{E}_{X \sim \Pi}[h(X_0, X_t)] = \mathbb{E}_{X \sim \Pi}[\mathbb{E}_{X \sim \Pi}[h(X_0, X_t) \mid X_0, X_1]]$$

We substitute the integral definition of the inner conditional expectation:

$$\begin{aligned}\mathbb{E}_{X \sim \Pi}[h(X_0, X_t) \mid X_0, X_1] &= \int_{\mathbb{R}^d} h(x_0, x_t) Q_{t|0,1}(\mathrm{d}x_t|x_0, x_1) \\ &= \mathbb{E}_{x_t \sim Q_{t|0,1}(\cdot|x_0, x_1)}[h(x_0, x_t)]\end{aligned}$$

Plugging this back into the outer expectation over the endpoints $(X_0, X_1) \sim \Pi_{0,1}$, we obtain the final tractable form:

$$\mathbb{E}_{(x_0, x_t) \sim \Pi_{0,t}}[h(x_0, x_t)] = \mathbb{E}_{(x_0, x_1) \sim \Pi_{0,1}} \left[\mathbb{E}_{x_t \sim Q_{t|0,1}(\cdot|x_0, x_1)}[h(x_0, x_t)] \right]$$

□

With that result we can write the minimization objective as follows (by discarding the scaling of $1/\sigma^2$):

$$\mathbb{E}_{(x_0, x_1) \sim \Pi_{0,1}} \mathbb{E}_{x_t \sim Q_{t|0,1}(\cdot|x_0, x_1)} \left[\left\| \frac{x_0 - x_t}{t} - u_\phi(t, x_t) \right\|^2 \right]$$

Furthermore because x_t is sampled from the Brownian Bridges we can write:

$$x_t = tx_1 + (1-t)x_0 + \sigma\sqrt{t(1-t)}\epsilon, \quad \epsilon \sim N(0, \mathbf{I})$$

If we substitute this result back to the objective we derive:

$$\mathbb{E}_{(x_0, x_1) \sim \Pi_{0,1}} \mathbb{E}_{x_t \sim Q_{t|0,1}(\cdot|x_0, x_1)} \left[\left\| x_0 - x_1 - \sigma\sqrt{\frac{1-t}{t}}\epsilon - u_\phi(t, x_t) \right\|^2 \right]$$

With that result we derive the final form of the regression problem:

$$\phi^* = \operatorname{argmin}_\phi \mathbb{E}_{t \sim U[0,1]} \left[\frac{1}{\sigma^2} \mathbb{E}_{(x_0, x_1) \sim \Pi_{0,1}^{2n}} \mathbb{E}_{x_t \sim Q_{t|0,1}(\cdot|x_0, x_1)} \left[\left\| x_0 - x_1 - \sigma\sqrt{\frac{1-t}{t}}\epsilon - u_\phi(t, x_t) \right\|^2 \right] \right]$$

As the original [Diffusion Schrodinger Bridge Matching](#) paper, describe in Appendix H, this loss function needs a scaling factor to avoid the singularity at $t = 0$:

$$\lambda(t) = \left[1 + \frac{\sigma^2(1-t)}{t} \right]^{-1}$$

Step 2: Simulating to Evaluate the Projections

The second step of the algorithm dictates to simulate the following SDE:

$$dY_t = u_{\phi^*}(1-t, Y_t)dt + \sigma dB_t, \quad Y_0 \sim \pi_1$$

In the simulation we gather the samples $(Y_0, Y_1)_i \in \mathbb{N}$. By reversing the samples acquired, i.e., $(X_0, X_1)_i = (Y_1, Y_0)_i$ we calculated the marginal $M_{0,1}^{2n+1}$. In order to calculate the reciprocal projection Π^{2n+1} we just simply simulate those $(X_0, X_1)_i$ samples through the Brownian Bridge:

$$X_t = tX_1 + (1-t)X_0 + \sigma\sqrt{t(1-t)}\epsilon$$

Step 3: Solving the Forward Regression Problem

This step dictates to solve the following regression problem:

$$\theta^* = \operatorname{argmin}_{\theta} \left\{ \mathbb{E}_{t \sim U[0,1]} \left[\frac{1}{\sigma_t^2} \mathbb{E}_{(x_t, x_1) \sim \Pi_{t,1}^{2n+1}} \|\sigma_t^2 \nabla \log Q_{1|t}(x_1|x_t) - u_{\theta}(t, x_t)\|^2 \right] \right\}$$

Following the same arguments as step 1 we derive the following regression problem:

$$\theta^* = \operatorname{argmin}_{\theta} \mathbb{E}_{t \sim U[0,1]} \left[\frac{1}{\sigma^2} \mathbb{E}_{(x_0, x_1) \sim \Pi_{0,1}^{2n+1}} \mathbb{E}_{x_t \sim Q_{t|0,1}(\cdot|x_0, x_1)} \left[\left\| x_1 - x_0 - \sigma \sqrt{\frac{t}{1-t}} \epsilon - u_{\theta}(t, x_t) \right\|^2 \right] \right]$$

Which requires the following scaling factor:

$$\lambda(t) = \left[1 + \frac{\sigma^2 t}{1-t} \right]^{-1}$$

Step 3: Simulating to Evaluate the Projections

The final step dictates to simulate the following SDE:

$$dX_t = u_{\theta^*}(t, X_t)dt + \sigma dB_t, \quad X_0 \sim \pi_0$$

We simulate it to gather the samples $(X_0, X_1)_{i \in \mathbb{N}}$, which correspond to $M^{2n+2}_{0,1}$. Finally to prepare for the next IMF iteration we need to calculate Π^{2n+2} which we find by passing those $(X_0, X_1)_i$ samples through the Brownian Bridge, like we did in step 2:

$$X_t = tX_1 + (1-t)X_0 + \sigma\sqrt{t(1-t)}\epsilon$$